Evaluate CRAFT + TrOCR on IAM Handwriting Dataset

Benchmarks CER and WER on Teklia/IAM-line (all splits) using the traditional OCR pipeline: **CRAFT detection → TrOCR recognition**.

> Runtime → Change runtime type → **T4 GPU** (recommended)

In [ ]:

In [1]:
```
# Install dependencies (Colab)
!pip install -q torch torchvision jiwer datasets transformers accelerate opencv-python-headless scikit-imag
```

In [2]:
```
# (Optional) Mount Drive and cache Hugging Face weights so they don't re-download each session
from google.colab import drive
import os

drive.mount('/content/drive')
os.environ['HF_HOME'] = '/content/drive/MyDrive/hf_cache'
print('HF cache:', os.environ['HF_HOME'])
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive",
force_remount=True).
HF cache: /content/drive/MyDrive/hf_cache
```

In [3]:
```
# (Optional) If you have a Git URL for this repo, you can clone it into /content.
# Leave REPO_GIT_URL empty to skip.
REPO_GIT_URL = ''
REPO_DIR = '/content/Image-of-handwriting-to-docx'

import os
from pathlib import Path

if REPO_GIT_URL:
    if not Path(REPO_DIR).exists():
        !git clone {REPO_GIT_URL} {REPO_DIR}
    else:
        print('Repo already exists at:', REPO_DIR)
```

```
else:
    print('Skipping clone (REPO_GIT_URL is empty).')
```

```
Skipping clone (REPO_GIT_URL is empty).
```

In [4]:

```
# Point this to the repo folder location in your Colab runtime.
# Options:
# 1) If you copied the repo into Drive: set REPO_DIR to that path.
# 2) If you uploaded a zip and extracted it: set REPO_DIR to the extracted folder.
#
# Example (Drive):
REPO_DIR = '/content/drive/MyDrive/cogs181/Image-of-handwriting-to-docx'


import os
from pathlib import Path

repo_path = Path(REPO_DIR)
if not repo_path.exists():
    raise FileNotFoundError(f'Repo folder not found: {repo_path}. Update REPO_DIR above.')

%cd {repo_path}
print('Working directory:', os.getcwd())
```

```
/content/drive/MyDrive/cogs181/Image-of-handwriting-to-docx
Working directory: /content/drive/MyDrive/cogs181/Image-of-handwriting-to-docx
```

In [9]:

```
# Run evaluation twice (0 = all samples).
#
# Mode A: CRAFT detection + crop + TrOCR
!python evaluate_ocr.py --use_craft True  --cuda True --trocr_device cuda --num_samples 0 --output_dir Eva
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rat
e limits and faster downloads.
The image processor of type `ViTImageProcessor` is now loaded as a fast processor by default, even if the m
odel checkpoint was saved with a slow processor. This is a breaking change and may produce slightly differe
nt outputs. To continue using the slow processor, instantiate this class with `use_fast=False`.
Loading weights: 100% 635/635 [00:00<00:00, 787.42it/s, Materializing param=encoder.layernorm.weight]
The tied weights mapping and config for this model specifies to tie decoder.model.decoder.embed_tokens.weig
ht to decoder.output_projection.weight, but both are present in the checkpoints, so we will NOT tie them. Y
ou should update the config with `tie_word_embeddings=False` to silence this warning
VisionEncoderDecoderModel LOAD REPORT from: microsoft/trocr-large-handwritten
Key                          | Status  |
-----------------------------+---------+-
encoder.pooler.dense.bias    | MISSING |
encoder.pooler.dense.weight  | MISSING |

Notes:
- MISSING       :those params were newly initialized because missing from the checkpoint. Consider training
on your downstream task.
Loading Teklia/IAM-line dataset (all splits) ...
Total samples in dataset: 10373
Evaluating on 10373 samples
  [1/10373] GT: put down a resolution on the subject  |  PRED: put down veso . union . on the subject
  [10/10373] GT: meeting of Labour 0M Ps tommorow . Mr. Michael  |  PRED: meet . fing Labour . 0"11 . ps .
feu mbpton . Mr.
  [20/10373] GT: that the House of Lords should be abolished  |  PRED: that the House . Lords . should . be
ey . bo . lis
  [30/10373] GT: that Labour should not take any steps  |  PRED: that Labour should not take any steps .
  [40/10373] GT: that the House of Lords should be abolished and  |  PRED: that the House . of lords . shou
ld . be abolished
  [50/10373] GT: Labour has to have an adequate number of  |  PRED: Labour has fo . have on adequate . numb
er .
  [60/10373] GT: the Federal Premier .  |  PRED: the Federal Pnennes
  [70/10373] GT: have an adequate number of members . THE  |  PRED: ha by on ad equal . te number of member
s . THE
  [80/10373] GT: ( 280,000 members ) and Mr. Harry Nkumbula's  |  PRED: 2.80 . 00 0 members . ) and mr . Ha
rry Nkumbula
  [90/10373] GT: getting an elected majority in Northern  |  PRED: getting . an elected . majority . in Nor
thern zhoi
  [100/10373] GT: the talks fall through . There are bound  |  PRED: the talles . fall . thro ugh . there a
re bound
  [110/10373] GT: Sir Roy's United Federal Party is  |  PRED: sir . Roy 2 . hui . ted . Federal . Party . n
o
```

[120/10373] GT: the London talks on the Protectorate's | PRED: the London talks . on the Protectorate 8
[130/10373] GT: are studying them today . The conference | PRED: are stud . y in g . them . today . The conference
[140/10373] GT: Africans the overall majority they are | PRED: Afric . cens . the over all . majori . t y . they q
[150/10373] GT: plans do not give the Africans the overall majo- | PRED: plans . clu . not give . the A fricans . the overal
[160/10373] GT: But representatives of Sir Roy Welensky , Prime | PRED: But representa . fives . of Sir . Roy . Welens . l
[170/10373] GT: the United Federal Party and the Dominion | PRED: f he United Federal Party . and the D ominion
[180/10373] GT: been secret negotiations on the Northern Rhodesia' | PRED: been secret . negotia . fy . oms . on the Northern
[190/10373] GT: Government . Mr. James Callaghan , Labour's Coloni | PRED: Government . Mr. James Lalla ghan , Labour's Coloni
[200/10373] GT: Northern Rhodesia is a member of the Federation . | PRED: Northern Rhodesia cy member o f the Federali . on
[210/10373] GT: Labour's Colonial spokesman , said Sir Roy had | PRED: la bour's . Coloni ol spo . keis man . said . Sir R
[220/10373] GT: Mr. Macleod thought the two Rhodesian | PRED: My ma cleod . thought . the two Rhodesian
[230/10373] GT: called on Mr. Macmillan to cease his negotiations | PRED: called . on Mr. Mac. ml hun . (ease . his . negul
[240/10373] GT: and State housing projects . | PRED: and . state . hansung pray ec. fs .
[250/10373] GT: his Housing Minister . It has aroused strong | PRED: his Housing Minister . it has arou sed shecurry .
[260/10373] GT: in Government and State housing projects . | PRED: ph bovervener . and State . housing projects .
[270/10373] GT: Robertson later disclosed he had sent a letter | PRED: Robertson . later . informed he had sent a feltie
[280/10373] GT: Robertson later disclosed he had sent a letter | PRED: Robert . you later . disclosed . he had sent o. le
[290/10373] GT: complaints about Weaver's loyalty . | PRED: complaints about men . ney o loyality .
[300/10373] GT: He said these concerned Mr. Weaver's | PRED: the said . these concerned . Mr. Weaver . s .
[310/10373] GT: he would perform " outstanding service " | PRED: he would per . farm . ful out s h . an ding zej ui
[320/10373] GT: nomination before it can be considered by the | PRED: nomination . before . nl . dm . h e considered . by
[330/10373] GT: over West Germany's cash offer to help | PRED: over wut . Germany . s cash offer to hel p
[340/10373] GT: PRESIDENT KENNEDY is ready to get tough | PRED: PRESIDENT ke N.N.EDY. is neady to get t hough

```
  [10290/10373] GT: coronary . And in that second  |  PRED: coronary studies in that second
  [10300/10373] GT: when you reach your parents'  |  PRED: when you reach your parents '
  [10310/10373] GT: fided in her , given some expla-  |  PRED: ficked her , given some expla-
  [10320/10373] GT: She went back to Philip . But  |  PRED: the went back to Philip . But
  [10330/10373] GT: little joy , or satisfaction , it had  |  PRED: little joy , or satisfaction it had
  [10340/10373] GT: to see what comes out at the inquest .  |  PRED: to see what comes out at the inquiry
  [10350/10373] GT: he 'll confine himself to the nervous  |  PRED: he'll confine himself to the nervous
  [10360/10373] GT: patronizing word , ' she replied . ' I despise  |  PRED: patronizing word ; she replied
. ' Despite
  [10370/10373] GT: the fake friend of the family , like the  |  PRED: the father friend of the family , li
ke the

==================================================
EVALUATION RESULTS
==================================================
CRAFT enabled   : False (TrOCR full-image)
TrOCR model     : microsoft/trocr-large-handwritten
Samples         : 10373
WER             : 0.0363 (3.63%)
CER             : 0.0121 (1.21%)
Time            : 667.7s (0.06s/sample)
Results CSV     : Evaluation_TrOCR_full/eval_results.csv
==================================================
Summary saved   : Evaluation_TrOCR_full/eval_summary.txt
```

In [11]:

```python
# Preview per-sample results (both modes)
import pandas as pd

pd.set_option('display.max_colwidth', 80)

for out_dir in ["Evaluation_CRAFT_TrOCR", "Evaluation_TrOCR_full"]:
    print("=" * 60)
    print(out_dir)
    df = pd.read_csv(f"{out_dir}/eval_results.csv")
    print(f"Mean WER: {df['sample_wer'].astype(float).mean():.4f}")
    print(f"Mean CER: {df['sample_cer'].astype(float).mean():.4f}")
    display(df.head(20))
```

```
============================================================
Evaluation_CRAFT_TrOCR
Mean WER: 0.8448
Mean CER: 0.2942
```

	index	ground_truth	prediction	sample_wer	sample_cer
0	0	put down a resolution on the subject	put down veso . union . on the subject	0.5714	0.2500
1	1	and he is to be backed by Mr. Will	and he 40 to be backed by Mr. will	0.2222	0.0882
2	2	nominating any more Labour life Peers	nomi . nati . ung . any more . Labour life . peers .	1.6667	0.4324
3	3	M Ps tomorrow. Mr. Michael Foot has	MPs . tomorrow Mr. Michael Foot has	0.4286	0.1143
4	4	Griffiths, M P for Manchester Exchange .	Griffi . ths . up P. for Manchester Exchange	1.0000	0.2500
5	5	is to be made at a meeting of Labour	is to he made af a meeting af Labour	0.3333	0.0833
6	6	A MOVE to stop Mr. Gaitskell from	a move . to stop . Mr. Gaits . kell . from	1.1429	0.4242
7	7	0M P for Manchester Exchange .	vim . kinche . slei . Exchange	1.0000	0.6000
8	8	A MOVE to stop Mr. Gaitskell from nominating	A move . fe stop . bahnkell . from no-mmature .	1.0000	0.5455
9	9	meeting of Labour 0M Ps tommorow . Mr. Michael	meet . fing Labour . 0"11 . ps . feu mbpton . Mr. Michael	1.1111	0.5435
10	10	any more Labour life Peers is to be made at a	hny more . la bour peens . fe he made . at n .	1.0000	0.4889
11	11	Foot has put down a resolution on the subject	foot . hon . puel . down . lumb . bution . om . the web . ject .	1.8889	0.7111
12	12	and he is to be backed by Mr. Will Griffiths ,	cuel . he 1) feo . he beached . will Griff thy	1.0909	0.5652
13	13	which would appear to " prop up " an out-dated	rw . hics . in two and old . appear . to prop a 01 out . -dated .	1.7000	0.7391
14	14	Though they may gather some Left-wing support , a	though they may gather . some left . using support ,	0.6667	0.2245
15	15	life peers into existence , they should not	piers . into . existe . are the ey . of trouled . not	1.5000	0.6512
16	16	and that Labour should not take any steps	and that Labour should . n. of take ann .	0.6250	0.2683
17	17	now put forward nominees . He believes	froms . pen for rward . # minies . He believes .	1.2857	0.4474

	index	ground_truth	prediction	sample_wer	sample_cer
18	18	large majority of Labour M Ps are likely to	arge . majority .ify Labour MPs . are likely to	0.5556	0.2093
19	19	that the House of Lords should be abolished	that the House . Lords . should . be ey . bo . lished .	1.1250	0.3721

```
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Evaluation_TrOCR_full
Mean WER: 0.0371
Mean CER: 0.0134
```

	index	ground_truth	prediction	sample_wer	sample_cer
0	0	put down a resolution on the subject	put down a resolution on the subject	0.0000	0.0000
1	1	and he is to be backed by Mr. Will	and he is to be backed by Mr. Will	0.0000	0.0000
2	2	nominating any more Labour life Peers	nominating any more Labour life Peers	0.0000	0.0000
3	3	M Ps tomorrow. Mr. Michael Foot has	MPs tomorrow . Mr. Michael Foot has	0.4286	0.0571
4	4	Griffiths, M P for Manchester Exchange .	Griffiths , MP for Manchester Exchange .	0.4286	0.0500
5	5	is to be made at a meeting of Labour	is to be made at a meeting of Labour	0.0000	0.0000
6	6	A MOVE to stop Mr. Gaitskell from	# MOVE to stop Mr. Gaitskell from	0.1429	0.0303
7	7	0M P for Manchester Exchange .	UMIP for Manchester Exchange .	0.3333	0.0667
8	8	A MOVE to stop Mr. Gaitskell from nominating	A MOVE to stop Mr. Gaitskell from nominating	0.0000	0.0000
9	9	meeting of Labour 0M Ps tommorow . Mr. Michael	meeting of Labour 0MPs tomorrow . Mr. Michael	0.3333	0.0652
10	10	any more Labour life Peers is to be made at a	any more Labour life Peers is to be made at a	0.0000	0.0000
11	11	Foot has put down a resolution on the subject	Foot has put down a resolution on the subject	0.0000	0.0000
12	12	and he is to be backed by Mr. Will Griffiths ,	and he is to be backed by Mr. Will Griffiths ,	0.0000	0.0000
13	13	which would appear to " prop up " an out-dated	which would appear to " prop up " an out-dated	0.0000	0.0000
14	14	Though they may gather some Left-wing support , a	Though they may gather some Left-wing support , a	0.0000	0.0000
15	15	life peers into existence , they should not	life peers into existence , they should not	0.0000	0.0000
16	16	and that Labour should not take any steps	and that Labour should not take any steps	0.0000	0.0000
17	17	now put forward nominees . He believes	now put forward nominees . He believes	0.0000	0.0000
18	18	large majority of Labour M Ps are likely to	large majority of Labour MPs are likely to	0.2222	0.0233

	index	ground_truth	prediction	sample_wer	sample_cer
19	19	that the House of Lords should be abolished	that the House of Lords should be abolished	0.0000	0.0000

In [12]:
```
# Print summaries (both modes)
for out_dir in ["Evaluation_CRAFT_TrOCR", "Evaluation_TrOCR_full"]:
    print("=" * 60)
    print(out_dir)
    with open(f"{out_dir}/eval_summary.txt") as f:
        print(f.read())
```

```
============================================================
Evaluation_CRAFT_TrOCR
CRAFT enabled: True
CRAFT model: /content/drive/MyDrive/cogs181/Image-of-handwriting-to-docx/CRAFT-pytorch/weights/craft_mlt_25
k.pth
TrOCR model: microsoft/trocr-large-handwritten
Samples: 10373
WER: 0.8240 (82.40%)
CER: 0.2927 (29.27%)
Time: 1371.1s

============================================================
Evaluation_TrOCR_full
CRAFT enabled: False
TrOCR model: microsoft/trocr-large-handwritten
Samples: 10373
WER: 0.0363 (3.63%)
CER: 0.0121 (1.21%)
Time: 667.7s
```

In [8]:
```
# (Optional) Copy results to Google Drive
import os
import shutil
from google.colab import drive

drive.mount('/content/drive')

for out_dir in ["Evaluation_CRAFT_TrOCR", "Evaluation_TrOCR_full"]:
```

```
    drive_dest = f"/content/drive/MyDrive/{out_dir}"
    if os.path.exists(drive_dest):
        shutil.rmtree(drive_dest)
    shutil.copytree(out_dir, drive_dest)
    print(f"Results saved to Google Drive: {drive_dest}")
```

```
Drive already mounted at /content/drive; to attempt to forcibly remount, call drive.mount("/content/drive",
force_remount=True).
Results saved to Google Drive: /content/drive/MyDrive/Evaluation_CRAFT_TrOCR
Results saved to Google Drive: /content/drive/MyDrive/Evaluation_TrOCR_full
```